

BINHAO WANG

Earthquake cycle simulations | Laboratory rock experiments | Observational seismology
Department of Earth Sciences | University of Southern California | binhao.wang@usc.edu

EDUCATION

University of Southern California

- PhD candidate, *Department of Earth Sciences*
- Advisor: Dr. Sylvain Barbot

Los Angeles, CA, US
2021-June 2026 (expected)

Peking University

- B.S., Geophysics, *School of Earth and Space Science*
- Merit Students Honor
- Advisor: Dr. Han Yue

Beijing, China
2017-2021

PUBLICATIONS

Wang, B., & Barbot, S. (2024). Rupture segmentation on the East Anatolian fault (Turkey) controlled by along-strike variations in long-term slip rates in a structurally complex fault system. *Geology*, 52(10), 779-783. <https://doi.org/10.1130/G52403.1>

Wang, B., & Barbot, S. (2023). Pulse-like ruptures, seismic swarms, and tremorgenic slow-slip events with thermally activated friction. *Earth and Planetary Science Letters*, 603, 117983. <https://doi.org/10.1016/j.epsl.2022.117983>

Zhang, S., Houston, H., **Wang, B.**, & Zhang, H. (2026). Mapping of absolute stresses around two California earthquakes reveals a very weak crust. *Earth and Planetary Science Letters*, 682, 119972. <https://doi.org/10.1016/j.epsl.2026.119972>

Liu, M., **Wang, B.**, Guvercin, S., Li, Z., Wang, T., Liu, C., Ji, L., Barbot, S. (2026). Dynamic segmentation of the Sagaing fault. *Science*. [DOI:10.1126/science.ady3237](https://doi.org/10.1126/science.ady3237)

Zhang, S., Houston, H., Huang, S., **Wang, B.** (under review), Favorable tidal stress triggers both more energetic and more abundant tremors

Wang, B., & Houston, H. (under review), Evolution of energy-to-moment ratio during earthquake ruptures.

Wang, B., Wu, B., & Zhou, Y. (under review), Global repeating earthquakes reveal a scattered recurrence time-moment scaling consistent with crack models in a complex fault zone. [preprint](#).

Wang, B., & Barbot, B. (under review), Reversal of frictional healing by hydrothermal alteration of contact quality.

Shiller, M., & **Wang, B.** (under review), The Influence of Ancient Mediterranean Earthquakes on their Written Records: A Robust Catalog Built through an Interdisciplinary Case Study of the Roman Historian Tacitus.

Liu, M., **Wang, B.**, Wu, B., Guvercin, S., Zhang, L., Yang, Z., Seyler, C., Platt, J., & Barbot, S. (under review), A rock-centric framework for earthquake dynamics.

ONGOING PROJECTS

Wang, B., Liu, M., & Barbot, S. (in prep), Simulating Fully-dynamic Earthquake Cycles Using Lab-derived, Physics-based Friction with Multiple Deformation and Healing Mechanisms.

Wang, B., Barbot, S., & Kitajima, H. (in prep), Negative healing in slide-hold-slide experiments explained by contact quality as a second state variable.

TEACHING EXPERIENCE

(9 semesters, 6 courses at USC, 1 at PKU, both grad and undergrad levels)

- GEOL440 Geophysics and Geoengineering, USC** 2022 Spring, 2024 Spring
- Heat, elasticity, fluids in Earth's lithosphere for senior undergraduates. Solve differential equations in geodynamics analytically and numerically. Visualizing geodata using GMT and MATLAB/Python.
- GEOL425 Data Analysis in Earth & Environmental Sciences, USC** 2023 Fall, 2025 Fall
- Statistics, spectral analysis, visualization in Python for senior undergraduates and graduate students.
- GEOL241 Energy Systems, USC** 2025 Spring
- ENST150 Environmental Issues in Societies, USC** 2024 Fall
- Lead discussions on environmental topics such as marine protected areas, deforestation.
- GEOL108 Crises of a Planet, USC** 2023 Spring
- GEOL105 Planet Earth, USC** 2022 Fall
- General education class. Lead lab experiments, hold office hours, grade assignments and exams.
- Dimensional Analysis and Back-of-Envelope Calculations in Geophysics, PKU** 2021 Spring

GRANT & AWARDS

- Green Foundation Postdoctoral Scholarship at UCSD 2026
- USC Graduate School Summer Research and Writing Grants (\$2,000) 2026
- USC PhD Achievement Award Nomination (1 per department) 2026
- USC Earth Sciences Graduate Student Research Award (\$3,000) 2025-2026
- Statewide California Earthquake Center (SCEC) Student Travel Award 2024, 2025
- SCEC Community Geological Framework Model and Thermal Model workshop Travel Award 2025
- Serpentine, Fluids and Plate Boundary Deformation in Subduction Zones workshop Travel Grant 2024
- Cargèse 2023 School on Subduction Zone Processes Travel Award 2023
- USC Dornsife Ph.D. Academy Scholarship and Research Fund (\$500) 2023
- Weiming Scholar (PKU Junior Scholar, Top 1%) 2021
- Peking University Merit Student Honor 2021
- Peking University 3rd Place Scholarship 2020
- Founder Scholarship, PKU 2019

SKILLS

- Programming: Python, Julia, MATLAB, Bash shell scripting, LaTeX
- Languages: Mandarin (native), English (fluent), Japanese (conversational)

LEADERSHIP EXPERIENCES & SERVICES

University of Southern California

Los Angeles, CA, US

Host of USC Lithosphere Dynamics Seminar

2023-2024

- Invite mainly early career speakers for the USC Lithosphere Dynamics Seminar series from diverse background and home institutes. Organize and host the seminar and speakers' on-campus visits. Organize student lunches with speakers.

Host of two talks of USC Earth Sciences Department Seminar

2025 Fall

Organizer of USC Earth Sciences Graduate Student Writing Groups

2025-2026

Graduate liaison for the Society of Earth Science Students (SESS)

2025-2026

SELECTED PRESENTATIONS

Wang, B., Barbot, S. D. (2026). Frictional Weakening by Hydrothermal Alteration of Contact Quality. **Oral** Presentation at *USC Lithosphere Dynamics Seminar*.

Wang, B., Liu, M. & Barbot, S. D. (2024, 2025). Simulating Earthquake Cycles Using Lab-derived, Physics-based Friction with Multiple Deformation and Healing Mechanisms. **Oral** Presentation at *2024 AGU, 2025 AGU, 2025 SCEC GFM/CTM/CRM workshop*.

Wang, B., Wu, B., & Zhou, Y. (2025), Quasi-periodic repeating earthquakes may follow normal 1/3 scaling laws. **Invited talk** at *Caltech Seismo-Lab Brown Bag Seminar*. **Poster** at *2025 AGU*. **Oral** at *2026 SSA*.

Wang, B., Liu, M. & Barbot, S. D. (2024). Effects of Competing Deformation and Healing Mechanisms in Earthquake Cycles. **Poster** Presentation at *2024 GRC Rock Deformation*.

Wang, B., & Barbot, S. D. (2025). Rupture segmentation on the East Anatolian Fault controlled by along-strike variations in slip rate. **Invited talk** at *Zhejiang University, Tongji University*.

Wang, B., & Barbot, S. D. (2023). Rupture segmentation on the East Anatolian Fault controlled by slip partitioning in a structurally complex fault system. **Oral** Presentation at *2023 AGU Annual Meeting (S22B-02)*.

Wang, B., & Houston, H. (2023). Rupture Characteristics of Large and Small Earthquakes - Insights from Moment Rate Functions. **Poster** Presentation at *Cargèse 2023 School on Subduction Zone Processes*.

Wang, B., & Barbot, S. D. (2023). Crack-like and pulse-like ruptures, seismic swarms, and tremorgenic slow-slip events in velocity-weakening, thermally-strengthening faults. **Invited talk** at *Institute of Geology, China Earthquake Administration*.

Wang, B., & Barbot, S. D. (2022). Pulse-like ruptures, seismic swarms, and tremorgenic slow-slip events with thermally activated friction. **Invited talk** at *Texas A&M University*.

Wang, B., & Barbot, S. D. (2021). Crack-like and pulse-like ruptures, seismic swarms, and tremorgenic slow-slip events in velocity-weakening, thermally-strengthening faults. **Oral** Presentation at *2021 AGU Annual Meeting*

LAB & FIELD EXPERIENCE

Texas A&M University, TX, US 2022

Over one month of friction experiments under hydrothermal conditions on triaxial apparatus on natural samples from the IODP around Alpine Fault, supervised by Prof. Hiroko Kitajima.

Southern San Andreas Fault (sSAF), CA, US 2025

Day trip to look at the creeping southern San Andreas Fault near Salton Sea, with Prof. Alexis Ault and Ashley Griffith.

Whipple Mountains, CA, US 2024

3 days, metamorphic core complex with Prof. John Platt

San Andreas Fault – San Gabriel Mountains, CA, US 2024

day trip, touch fault gouge, damage zone, paleoseismic trench with Prof. Jean-Philippe Avouac

Pachalka Springs, CA, US 2024

3 days, mapping, overturned folding of sedimentary sequence, exotic granite with Prof. John Platt

Catalina Island, California (CA), US 2023-2024

- (Jan 2024) One week trip with MIT geophysical group (Profs. William Frank, Brent Minchew), structural geology (Profs. John Platt, Sylvain Barbot), installing seismometers and gravity meters
- (2023 Spring) 3-day trip, exhumed subduction zone, metamorphic units and ductile shear zone with Prof. John Platt.

Inyo Mountains, Owens Valley, CA, US 2022

- 3-day trip to map the 1km x 1km region with a fold and two faults

Mt San Jacinto, Palm Springs, CA, US 2022

- Hunt for pseudotachylytes with Prof. Sylvain Barbot

Arroyo Burro, Santa Barbara, CA, US 2022

- Day trip to measure the fold axis of a fold

Northern Death Valley and Garlock fault, CA, US 2022

- 3-day trip to visit foldings, paleoseismology

Mojave Desert, CA, US 2022

- 3-day trip to visit Amboy Crater, lava tubes, Kelson Dunes.

Two Lights State Park, Cape Elizabeth, Maine, US 2024

- Day trip, Norumbega-Two Lights Fault Zone, pseudotachylyte, GRC2024 Rock Deformation workshop. Field guide: Prof. Emily Peterman.

last updated: Apr-13-2026

Daliangshan, Sichuan, China, seismic network

2020

- 15-day trip to employ short-period seismometers in Xiaojiang Fault Zone, east of the Tibet Plateau.